

Leadership Team – Aminyx Therapeutics



Matthew L. Becker, Ph.D.
Co-Founder



Thomas X. Neenan, Ph.D.
Co-Founder

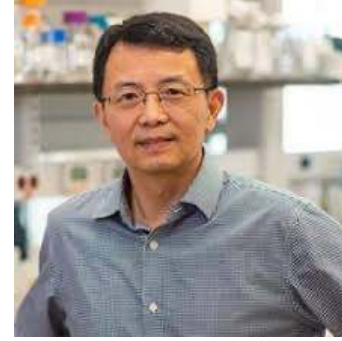
Co-Founder & CSO @ Panbela
Therapeutics (NASDAQ: PBLA)

Founder & CBO @ Nanite Bio

Founder & CEO @ AbFero



Courtney Dzievior, Ph.D.
Co-Founder



Ru-Rong Ji, Ph.D.
Scientific Advisory Board
Department of Anesthesiology
Duke University



Aaron Smith
Advisor
Former CEO, Artelon

Aminyx Therapeutics

Timing, Technology & Team are all Aligned

\$8.1 Billion total addressable market

Substantial Clinical Advantage: > 4-day pain relief

Commercialization within 3 years

Patented, disruptive CYPREX Poly(ester urea) technology

Social Impact Investment: Alternative to opioids

Management of Postoperative Pain Remains a Challenging Problem

Annual Surgeries

69 MILLION
in the U.S.

234 MILLION
Globally

65%

Will Experience

Moderate to Severe Postoperative Pain ¹

Suboptimal acute-postoperative pain management causes an array of negative consequences, including increased morbidity, impaired physical function and quality of life, slowed recovery, prolonged opioid use during and after hospitalization, and increased cost of care.²

Annual Prescriptions in U.S.

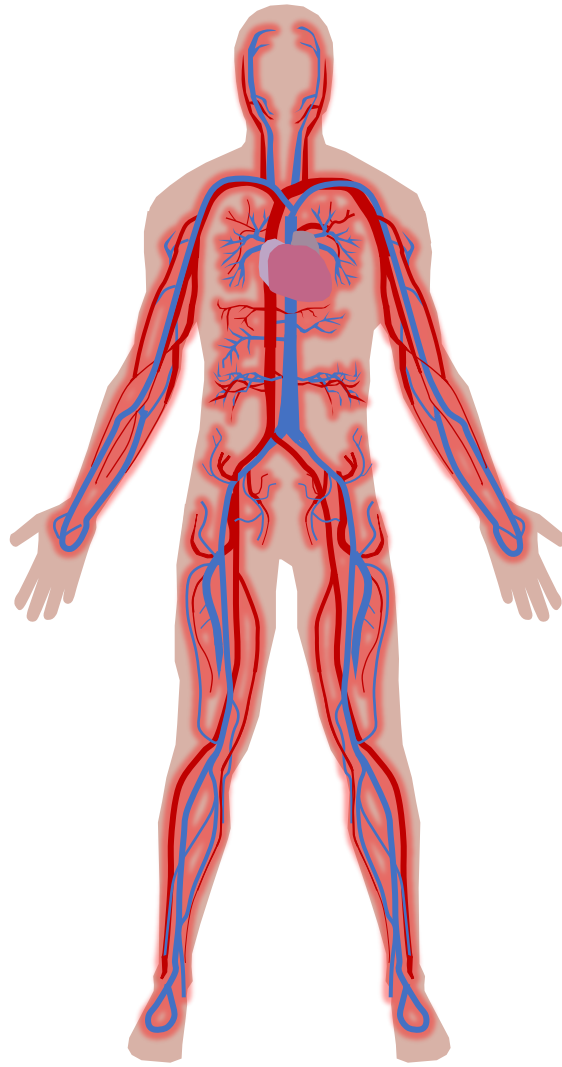
550
MILLION
Pain Medications

250
MILLION
Opioids

¹ Centers for Disease Control and Prevention, 2015; Weiser et al *Lancet*, 12;372(9633): 139-44; Symphony Health; United Nations Office on Drugs and Crime, 2015

² Gan T, Poorly controlled postoperative pain: prevalence, consequences, and prevention, *Journal of Pain Research* 2017; 10: 2287–2298.

Two Distinct Strategies for Drug Delivery : Systemic vs. Local



Systemic

Oral
Intravenous

Systemic delivery increases adverse risks

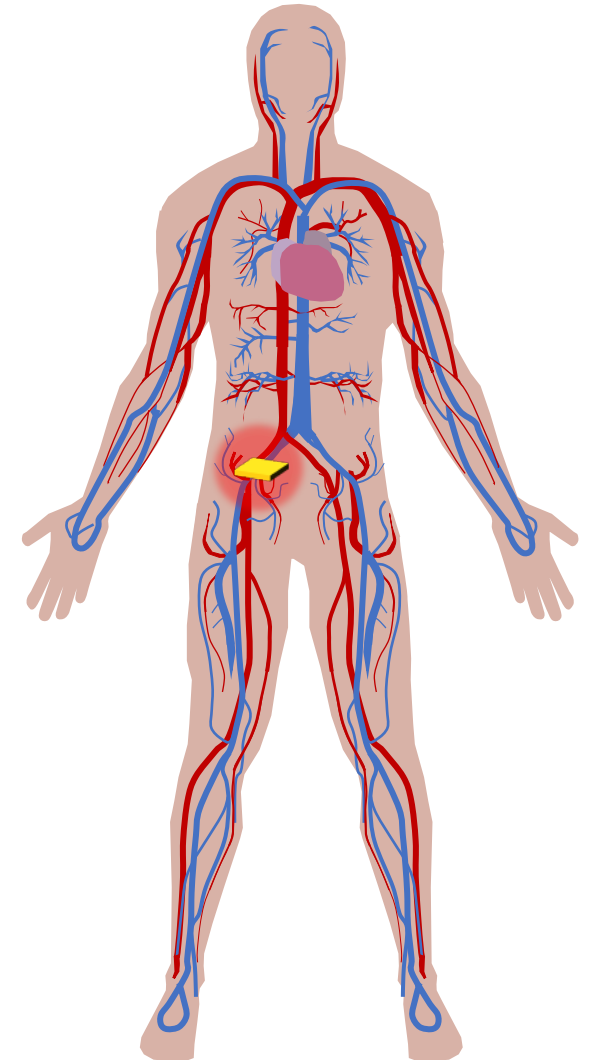
Healthy tissue also gets affected

Drug goes through biological changes systemically

Local delivery provides bioavailability direct to pain area

Local

Implantation
Some Injections



The Opportunity

The global pain management market size will grow to nearly USD 125.6 billion in 2034, with a CAGR of 4.10 %.

The market is fueled by rising demand for non-opioid alternatives, innovations in drug delivery systems, and increasing cases of chronic pain.

Aminyx Therapeutics Advantage

Strong Patent Portfolio

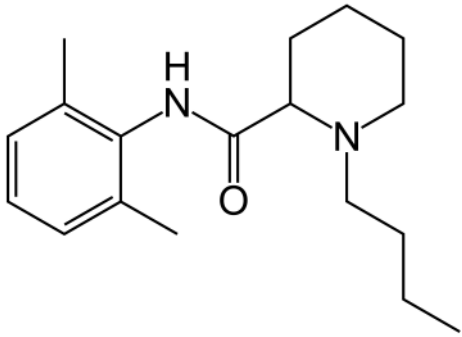
Extensive Published Peer-Reviewed Data

Large Animal pK & safety data

Regulatory path is very straight forward

Research Phase is Complete – Focused on development

Implementation of Bupivacaine & Meloxicam-based Therapies



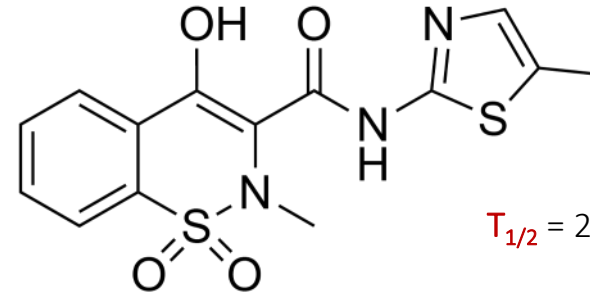
Cardiac toxicity at high doses

CD_{50} – 57.7 mg/kg (mice, intraperitoneally)¹

LD_{50} – 58.7 mg/kg (mice, intraperitoneally)¹
 18 mg/kg (rabbit, oral)²
 6 mg/kg (rat, para-periosteal)²
 43 mg/kg (rat, subcutaneous)²
 6.1 mg/kg (mice, intravenous)²

Long duration ($T_{1/2}$ = 2.7 hours) of action in comparison to other LAs

ED_{50} – 5.43 – 6.10 mg (human, intrathecal)³
*depends on patient and administration route



$T_{1/2}$ = 20 hours¹

LD_{50} – 470 mg/kg (mice, oral)²
 83.5 mg/kg (rat, oral)²
 320 mg/kg (rabbits, oral)²
 1600 mg/kg (mini-pigs, oral)²

FDA Approved Products:

ZYNRELEF

US approval: 2021
 Bupivacaine + meloxicam
 Injection

36 hours

XARACOL

US approval: 2020
 Bupivacaine HCl
 Inguinal Implantation

24 hours

EXPAREL

US approval: 2011
 Liposomal Bupivacaine
 Injection

< 30hours

MARCAINE

Bupivacaine HCl
 Injection

8 hours

SENSORCAINE

Bupivacaine HCl +/-
 epinephrine Injection

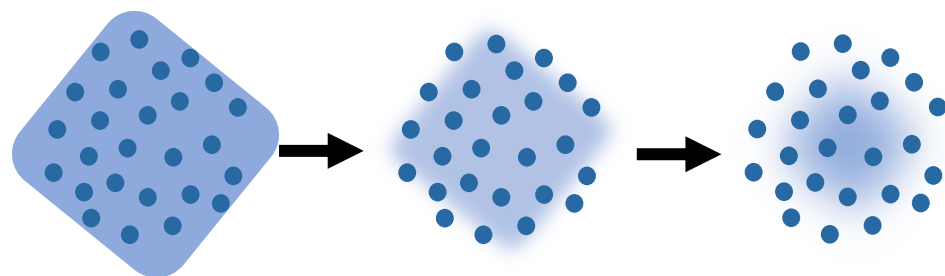
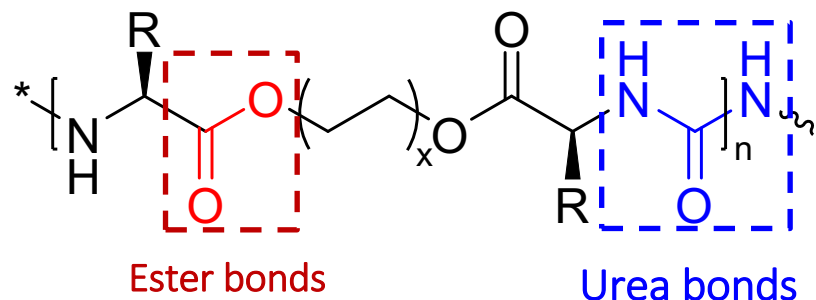
8 hours

¹ de Jong RH, Bonin JD. *Anesth Analg*. **1980**, 59(6), :401-5. PMID: 7189975.

² Material SDS. https://cdn.pfizer.com/pfizercom/products/material_safety_data/PZ00763.pdf

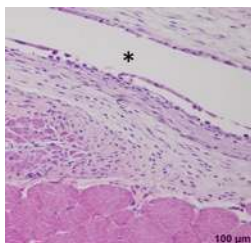
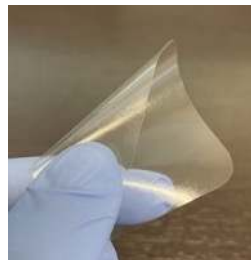
³ Chen MQ, Xia ZY. *Med Sci Monit*. **2015**, 21, 2588-94. doi: 10.12659/MSM.894842

Amino Acid Based Poly(ester urea)s



Uniform degradation &
Non-acidic byproducts

U.S. Patents: 9,988,492; 10,537,660; 10,280,261;
11,103,613 w/ others pending



Versatile thermoplastic

Many validated form factors

Film, tube, suture, mesh, scaffold

Synthetic Flexibility

Tunable mechanical properties

Customizable degradation (weeks to months)

Biocompatible

Non-toxic byproducts

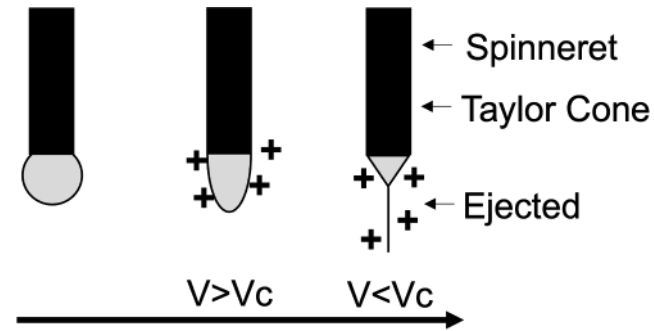
Limited inflammatory response

Controlled drug delivery

Drug stabilization

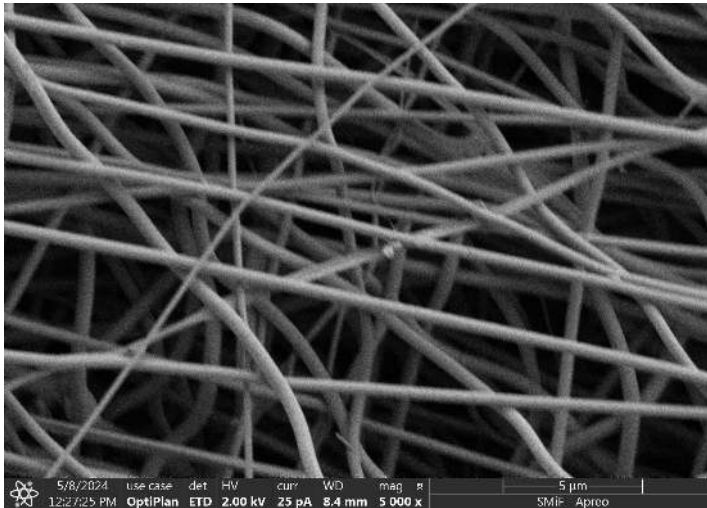
Uniform and sustained (days to weeks)

Electrospinning of Drug-loaded Poly(ester urea) Substrates



Nanofibers with variable fiber diameters can be fabricated by varying:

- 1) Needle to collector distance
- 2) Solution viscosity

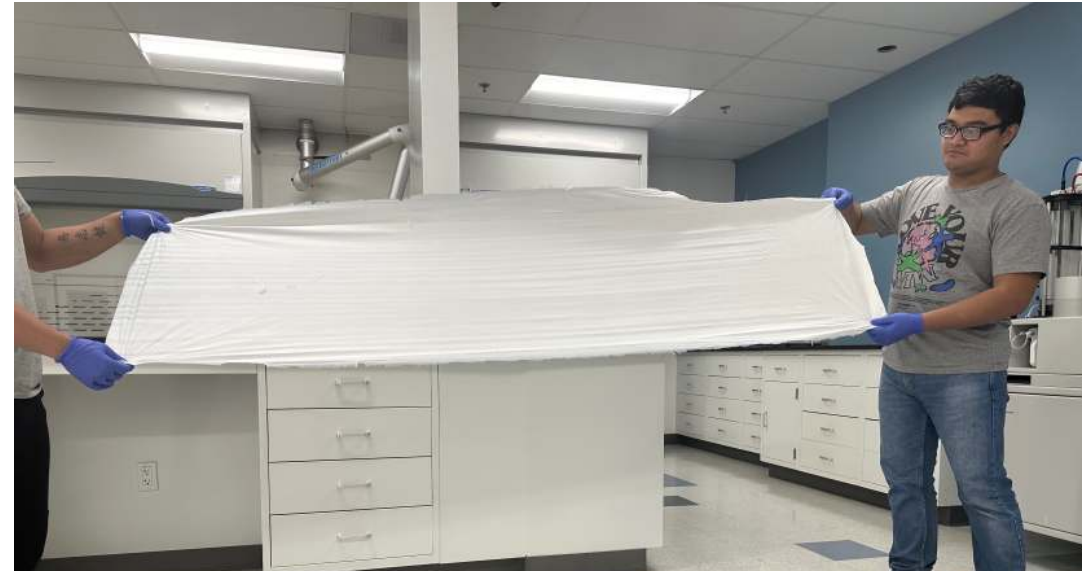


Very high API loading

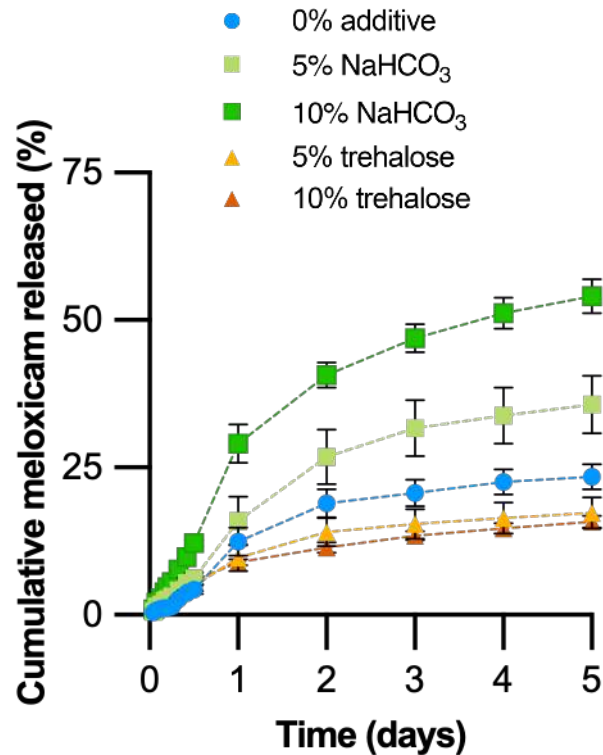
40% by mass Bupivacaine

20% by mass Meloxicam

Scalable

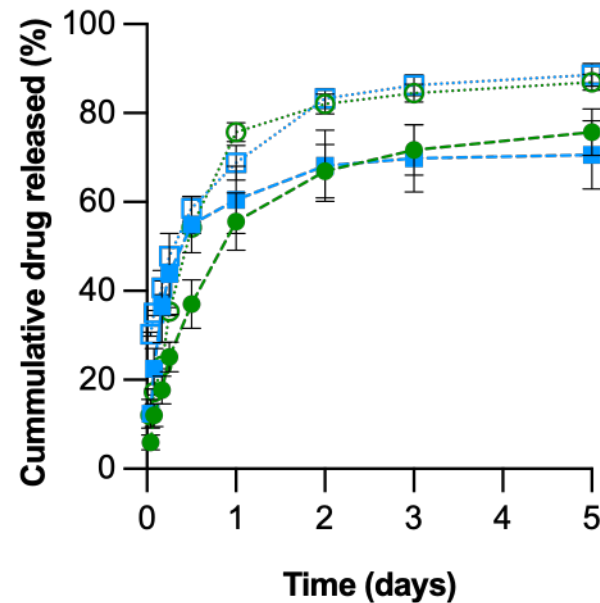


Development of Controlled-Release Pain Management Platform Outperforms Clinical Standards



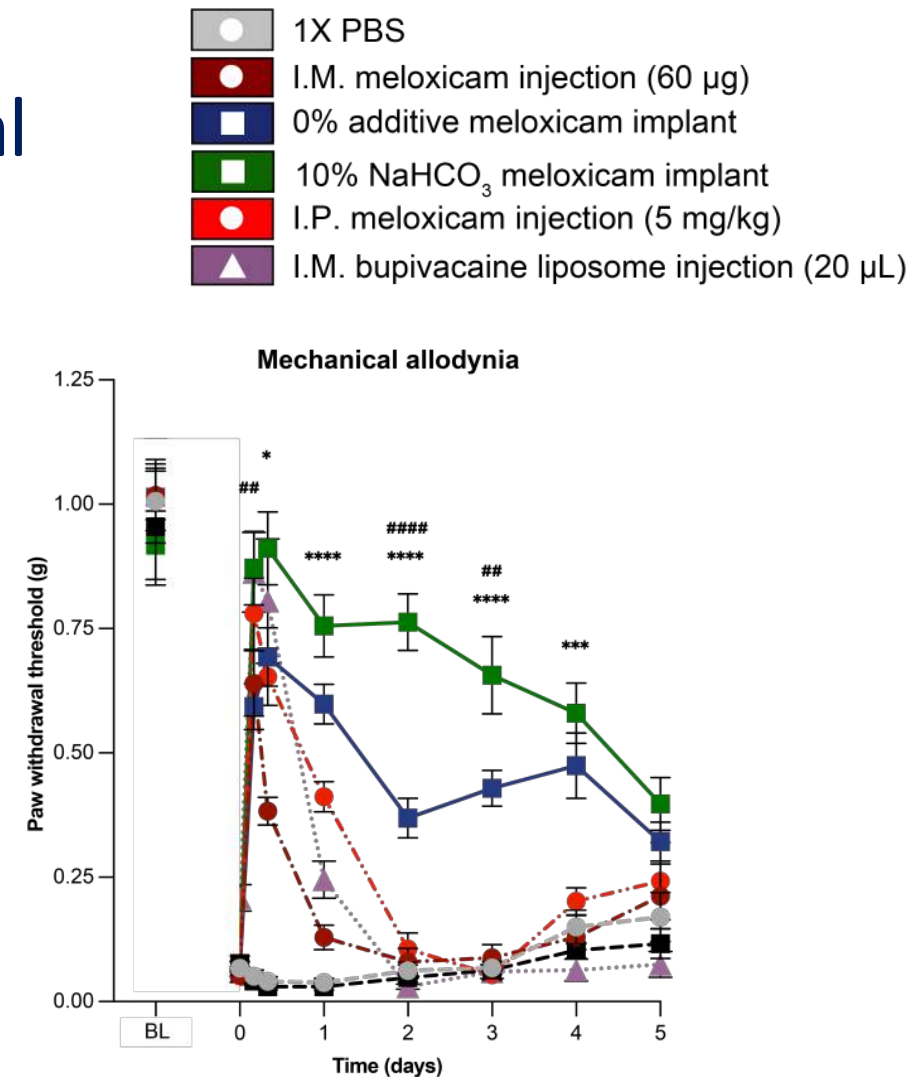
Optimized drug release

Varied formulation components, load, and thickness



Effective pain management

Superior performance over clinical standards



Translatable platform

Versatile across therapeutics and multimodal delivery strategies